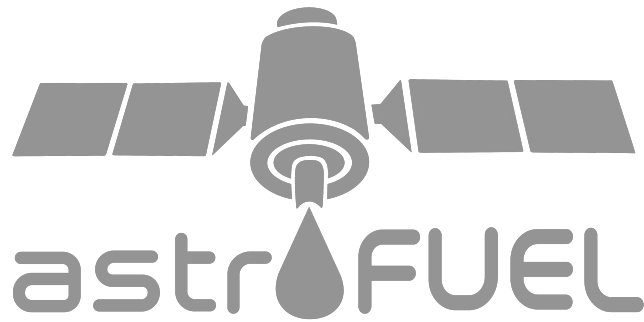




LAZARUS - REQUIREMENTS

ASTROFUEL

Authors: Cesseti, Federico
Gómez Jiménez, Paula
González Gómez, Víctor
Hernández de Rojas, Pelayo
Ramiro Rodríguez, Pablo
Sánchez Romero, Ana
Santoro, Stefano
Tarancón Cámara, Mario
Tozza, Antonio

MADRID, SEPTEMBER 2026

1. Scope of the mission

AstroFUEL focuses on developing an orbital refuelling infrastructure to extend the life of satellites in geostationary orbit (GEO). Its main goal is to offer an alternative for satellites that are otherwise fully functional but lack the needed propellant to maintain their orbit. By avoiding unnecessary replacements, the project aims to reduce space debris and cut costs, giving satellites a second life.

The first mission is called Lazarus, targeted for the 2030's market, inspired by the idea of bringing satellites back to life. It consists of a refuelling satellite that approaches a client satellite to provide the fuel it needs.

The satellite will carry the propellant, stored in high-pressure tanks. It will also have a standard refuelling interface and a docking mechanism compatible with future satellites. The satellite will be able to perform rendezvous and docking manoeuvres, transfer propellant to the client, and safely separate from it after refuelling. Once the operation is completed, Lazarus will be ready to repeat the process with other satellites.

2. Requirements

2.1. Mission Level Requirements

Mission Requirements	
Req. ID	Statement
MIS_010	The satellite shall be capable of providing refuelling services to different types of client satellites, regardless of their geometry, size and configuration, provided that they have a compatible refuelling interface.
MIS_020	The satellite shall have a mechanical interface compatible with the standard refuelling ports expected to be used by future client satellites, allowing a safe and stable connection during xenon transfer, as well as a controlled and safe undocking once the operation is completed.
MIS_030	The satellite shall be capable of performing the necessary orbital manoeuvres to reach and assist a client satellite in geostationary orbit (GEO) within a maximum of 10 hours after receiving an emergency service request.
MIS_040	The satellite shall have a minimum operational lifetime of 2 years after insertion into its nominal orbit, during which it shall maintain the capabilities required to perform the planned refuelling services.
MIS_050	The satellite subsystems shall be designed and sized considering the expected needs of the space market from the 2030s onwards, taking into account technological developments, the characteristics of client satellites and the operational requirements expected during the mission's service life.
MIS_060	The total mass of the satellite at launch (wet mass), including the structure, subsystems, xenon payload and propellant for its own manoeuvres, shall not exceed 1,500 kg.

Mission Requirements	
Req. ID	Statement
MIS_070	The satellite shall have a minimum payload capacity of 500 kg of xenon to supply different client satellites. As a reference scenario, five refuelling operations of 100 kg each are considered, with the aim of extending the operational lifetime of each client by approximately five years. However, the amount of propellant transferred during each operation may be adjusted according to the client's needs, allowing a different number of refuelling operations, provided that the total available payload capacity is not exceeded.
MIS_080	The satellite's tank shall ensure the safe storage of the xenon payload throughout the mission's operational lifetime, maintaining its structural integrity and preventing leaks under the conditions of the space environment.
MIS_090	The satellite shall be capable of performing autonomous rendezvous manoeuvres at any point in its orbit, maintaining communication with the client satellite throughout the process to ensure a coordinated and safe approach.
MIS_100	The satellite shall be capable of performing accurate and safe approach manoeuvres without entering GEO orbital slots assigned to other satellites. During the approach, it shall position itself at a relative distance of no more than 500 m from the client satellite, maintaining the necessary safety conditions to continue with docking and refuelling operations.
MIS_110	The satellite shall be capable of detecting unsafe conditions during approach or docking manoeuvres and automatically aborting the operation, performing the necessary manoeuvres to ensure a safe separation from the client satellite.
MIS_120	The satellite shall have a system capable of transferring xenon to the client satellite under vacuum and microgravity conditions, using the necessary connection, pressure regulation and flow control mechanisms to ensure a safe and controlled transfer, while respecting the operating pressure and temperature limits of both spacecraft.
MIS_130	The satellite shall maintain continuous coverage, except during maneuvers required to provide the docking service.

References

- [1] Bombardelli C. Astrodynamics and Attitude Dynamics: Gravity Assist and Flyby Dynamics. Madrid, Spain; 2023. Lecture notes, Master's Degree in Aeronautical Engineering.
- [2] Bombardelli C. Astrodynamics and Attitude Dynamics: The Two-Body Problem (Review). Madrid, Spain; 2025. Lecture notes, Master's Degree in Aeronautical Engineering.
- [3] del Cura Velayos JM. Systems Engineering and Preliminary Design of Space Vehicles. Topic 3: Preliminary Design of a Space Mission. Presentation. Madrid, Spain; 2020. Lecture notes.